

Proof. Theorem (Bernoulli's inequality). Let $x \in \mathbb{R}$ with $x \geq -1$ and let $n \in \mathbb{N}$ (i.e. $n \geq 0$). Then

$$(1+x)^n \geq 1+nx.$$

Base case $n = 0$.

$$(1+x)^0 = 1 = 1 + 0 \cdot x.$$

Hence the inequality holds with equality for every admissible x .

Inductive step. Assume that for some $n \geq 0$ and for every $x \geq -1$

$$(1+x)^n \geq 1+nx. \quad (\text{IH})$$

Because $x \geq -1$ we have $1+x \geq 0$, so multiplication by the non-negative factor $1+x$ preserves the inequality direction:

$$\begin{aligned} (1+x)^{n+1} &= (1+x)^n(1+x) \\ &\geq (1+nx)(1+x). \end{aligned} \quad (1)$$

Expanding the right-hand side gives

$$(1+nx)(1+x) = 1 + (n+1)x + nx^2. \quad (2)$$

Since $n \geq 0$ and $x^2 \geq 0$, the term nx^2 is non-negative. Dropping this non-negative term yields the weaker (hence still valid) inequality

$$1 + (n+1)x + nx^2 \geq 1 + (n+1)x. \quad (3)$$

Combining (1)–(3) we obtain

$$(1+x)^{n+1} \geq 1 + (n+1)x,$$

which is precisely the desired statement for the exponent $n+1$.

Conclusion. The base case $n = 0$ is true, and the inductive step shows that truth for a given n implies truth for $n+1$. By the principle of mathematical induction the inequality holds for every integer $n \geq 0$ and every real $x \geq -1$.

Equality cases. Equality can occur only in the steps where a non-negative term was discarded.

- In the base case $n = 0$ we have equality for all admissible x .
- In the inductive step the term nx^2 was removed. Equality in that step requires $nx^2 = 0$, i.e. either $n = 0$ (the base case) or $x = 0$.

Consequently the original inequality becomes an equality precisely in the following situations

$$\begin{cases} n = 0 & (\text{any } x \geq -1), \\ n = 1 & (\text{any } x \geq -1), \\ x = 0 & (\text{any } n \geq 0). \end{cases}$$

(For $x = -1$ the inequality reads $0 \geq 1 - n$, and equality holds only when $n = 1$, which is already included above.) Thus equality holds *iff* $x = 0$ or $n \leq 1$. □ □